Natural SNP-RFLP Primer Design using Memetic Algorithm

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Abstract

Single nucleotide polymorphisms (SNPs) can be genotyped by polymerase chain reaction-restriction fragment length polymorphism (PCR-RFLP). Before performing PCR-RFLP for SNP genotyping, a feasible primer pair observes numerous constraints and an available restriction enzyme discriminate the target SNP, are required, here we called it natural SNP-RFLP primer design. Design the natural SNP-RFLP primers is usually tedious and challenging problem. In order to get the available primers and restriction enzymes effectively, we use a memetic algorithm (MA) to cooperate with SNP-RFLPing. The test results shown it is able to design feasible primers that fit the common primer constraints and provide the available restriction enzymes.

1 Introduction

Single nucleotide polymorphism (SNP) genotyping plays an important role in many researches, such as forensics [1], malignancy studies [2, 3], personalized medicine [4] and population genetics and evolutionary studies [5]. Many laboratories use polymerase chain reaction-restriction fragment length polymorphism (PCR-RFLP) to investigate the causes of genetic variations and mutations. PCR-RFLP is considered especially useful in small-scale research studies of complex genetic diseases [6]. PCR-RFLP is also used to perform SNP genotyping [6, 7]. When performing SNP genotyping, a feasible primer pair is needed and a given target SNP is discriminated by natural digestion with specific restriction enzymes, here we called the process “Natural SNP-RFLP primer design”. In order to design a feasible primer pair observes numerous constraints and get an available restriction enzyme for discriminating the target SNP, several computer programs are developed.

For example, V-MitoSNP designs a SNP-RFLP primer set for mtSNPs [8]. SNP Cutter uses Primer3 [9] to look for SNP-RFLP primer sets. Prim-SNPing provides a natural SNP-RFLP primer design function for cost-effective SNP genotyping [10]. However, these tools are free of an effectively specialized method. Therefore, the development of natural SNP-RFLP primer design is still mandated.

Natural SNP-RFLP primer design is a tedious and challenging task since numerous primer constraints must be considered [9, 11-13] and available restriction enzymes need to be mined. Typical primer constraints include the primer length, primer length difference, PCR product size, GC proportion, melting temperature (T_m), melting temperature difference ($T_{m\text{-diff}}$), GC clamp, dimers (including cross-dimers and self-dimers), hairpin structure, and specificity. In the past, we had used a genetic algorithm (GA) to design the natural SNP-RFLP primers [14]. In this paper, we introduce the memetic algorithm (MA) [15] to facilitate the design for available primer sets [14], and cooperate with SNP-RFLPing [16-18] to mine available restriction enzymes from REBASE [19]. The following sections describe the proposed method and its test results.

2 Methods

Problem formulation

Let T_D be a DNA template sequence that is made up by nucleotide codes of the DNA with an identified SNP. T_D is defined as follows:

$$T_D = \{B_i \mid i \text{ is the index of DNA sequence}, \exists! B_i \in \text{SNP}\} \tag{1}$$

where B_i represents the regular nucleotides ‘A’, ‘T’, ‘C’, ‘G’, the SNP IUPAC code (M, R, W, S, Y, K, V, H, D, B or N) or the dNTPs format ([dNTP1/dNTP2]). The symbol $\exists!$ represents the existence and uniqueness. We focus only on true SNPs as described in dbSNP [20] of NCBI as the target SNPs, i.e., deletion/insertion polymorphisms (DIPs) and multi-nucleotide polymorphisms (MNPs) are not included.

The natural SNP-RFLP primer design problem now consists of finding a pair of sub-sequences of corresponding constraints from T_D that contains at least one restriction enzyme can distinguish a given target SNP. One sub-sequence is called the forward primer (P_f) and the other is called the reverse primer (P_r). The forward primer and the reverse primer are defined as follows:

$$P_f = \{B_i \mid \forall B_i \in \{'A','T','C','G'\}, F_s \le i \le F_e, i \text{ is the index of } T_D\} \quad (2)$$

$$P_r = \{\overline{B_i} \mid \forall B_i \in \{'A','T','C','G'\}, R_s \le i \le R_e, i \text{ is the index of } T_D\} \quad (3)$$

where F_s and F_e denote the start index and the end index of P_f in T_D; R_s and R_e denote the start index and the end index of P_r in T_D. P_f and P_r are called a primer pair. $\{\overline{B_i}\}$ is the anti-sense sequence of B_i. For example, a sequence B_i = "TACGTGTCCGAAC", its complement sequence is "ATGCACAGGCTTG", since the complement nucleotide of 'A' is 'T'; the complement nucleotide of 'C' is 'G', and vice versa. The anti-sense sequence $\{\overline{B_i}\}$ is the reverse of the complement sequence, i.e., $\{\overline{B_i}\}$ = "GTTCGGACACGTA".

Here, we define a simple vector (called "individual" in a MA) consisting of F_s, F_l, P_l and R_l (shown as Figure 1) that can determine a primer pair below.

$$P_v = (F_s, F_l, P_l, R_l) \quad (4)$$

Consequently, the forward primer and the reverse primer can be obtained from P_v. P_v is the encoding prototype in the natural SNP-RFLP primer design problem. In later sections, P_v is used to perform evolutionary computations of the MA.

Fitness function

In order to make a primer pair satisfy the design constraints, primer constraints are used to constitute a fitness function to evaluate its fitness value. The fitness value is minimized (i.e., a fitness value equal to 0 indicates the best natural SNP-RFLP primer pair). Eight primer constraint functions are included in the proposed fitness function (Eq. 5).

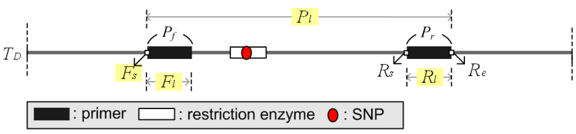

Figure 1. Diagram for the natural SNP-RFLP primer set.

$$\begin{aligned} Fitness(P_v) = {} & 3\times(Len_{diff}(P_v)+GC_{proportion}(P_v)+GC_{clamp}(P_v))+ \\ & 10\times(Tm(P_v)+Tm_{diff}(P_v)+dimer(P_v)+hairpin(P_v))+ \\ & 50\times specificity(P_v)+60\times product\,(P_v) \end{aligned} \quad (5)$$

We use weights to discriminate the significance of each primer constraint function by the experiences of an expert and the experimental requirements of the PCR experiments. Four different weight values represent different degrees of importance for these functions, namely 3, 10, 50 and 60. Users can make a fine tuning to satisfy their different experimental requirements. Larger weight value the more importance constraint function. These related primer constraints and functions are described below.

The primer length is set to between 16 and 28 nt. For longer primers, the T_m is increased, whereas the T_m of relatively short primers is decreased. Accordingly, neither primer that is too long nor too short is suitable. Since the random values of F_{l1}, R_{l1}, F_{l2} and R_{l2} are already limited by the primer length constraint, the primer length estimation is not considered into the fitness function. A length difference (Len_{diff}) of less than or equal to 3 nt between the F_l and the R_l is considered optimal. The $GC_{proportion}(P_v)$ function is used to check whether the $GC\%(P)$ of the forward and reverse primers is between 40% and 60% or not. To account for the presence of G or C at the 3' terminal of a primer, the function $GC_{clamp}(P_v)$ is proposed. Similar T_m values in a primer pair are important for a PCR experiment. The $Tm(P_v)$ function is used to check whether the melting temperature T_m of a primer pair lies between 45°C and 62°C, and the $Tm_{diff}(P_v)$ function checks whether the difference of the melting temperature exceeds 5°C. The $dimer(P_v)$ and $hairpin(P_v)$ functions check for the existence of dimers or hairpin structures in the designed primer pair. The $specificity(P_v)$ function is proposed to check for repetition of each primer in the template DNA sequence to ensure its specificity. Finally, we use $product\ (P_v)$ function to design the PCR product length more than 100 bp.

Algorithm

The proposed method contains six separate processes. These processes are 1) mining of SNP-RFLP restriction enzymes, 2) judgment on availability of restriction enzymes, 3) creation of a random initial population, 4) evaluation of the fitness, 5) local search, 6) judgment of termination criteria, 7) selection, crossover and mutation, and 8) replacement operation, respectively. These processes are described below.

1) Mining of SNP-RFLP restriction enzymes. First, the proposed method uses SNP-RFLPing [16-18] to mine the available restriction enzymes for discriminating a given target SNP. SNP-RFLPing was developed by our team in order to obtain available restriction enzymes.

2) Judgment on availability of restriction enzymes. If no restriction enzyme is available to distinguish the alleles of the given target SNP, the natural SNP-RFLP primer design is insignificant, and the proposed method is terminated. Otherwise, the proposed method performs the following processes.

3) Creation of a random initial population. A preset number of individuals (P_v) are randomly generated for an initial population without duplicates. The P_v is defined in the previous **Problem formulation** section.

4) Evaluation of the fitness. The fitness value in the fitness function is used to individually ascertain whether a chromosome is good or bad. We use fitness function (eq. 5) to evaluate the fitness values of all chromosomes in the population for related chromosomal operations later.

5) Local search. The local search identifies superior individuals from amongst the neighbors of the original individual. The experience of the original individual is improved, and thus a local optimum solution can be obtained. Each iteration, all offsprings are subjected to the local search process so that local optimality is always preserved. Finally, a global optimality is determined. The pseudo-code for the local search is shown below:

Local search pseudo-code

```
Begin;
  Select an incremental value d=a×Rand();
  For a given chromosome i∈P: calculate fitness(i);
  For j=1 to number of variables in chromosome i;
    value(j)=value(j)+d;
  If chromosome fitness does not improve then
    value(j)=value(j)–d;
  else if chromosome fitness improves then
    retain value(j);
  Next j;
  End;
```

In the local search, P represents a population and d is calculated to assist an individual in seeking out neighboring individuals. Before calculating d, a must be determined, which is a constant that suits the variable values. The details can refer to [15].

6) Judgment on termination conditions. Once an optimal solution of chromosomes or the maximum number of iterations is reached or the fitness value is 0, the MA is terminated.

7) Selection, crossover and mutation. In MAs, the processes for evolutionary computation are similar to GAs, including selection, crossover and mutation. We applied a random selection to select two chromosomes from the population. The selected two chromosomes are processed by either the crossover or mutation operation. Uniform crossover is used to implement the crossover operation. One-point mutation is applied to the mutation operation.

8) Replacement operation. After the evolutionary computation processes have been implemented, the two worst chromosomes in a population are replaced by the new offsprings, and the process is repeated in the next generation until the termination condition is reached.

3 Results

The *in silico* environment

The proposed method was run on a Xeon(TM) CPU 3.20 GHz × 2 and 2GB RAM under the Microsoft Windows XP SP3 and JAVA 6.0 platforms.

Test template sequences

A point mutation in the SLC6A4 gene was identified and shown to be associated with psychosis [21], bipolarity [22], and autism spectrum disorders [23] in patients. The SLC6A4 gene is the member 4 for a solute carrier family 6 (neurotransmitter transporter, serotonin). Overall, 288 SNPs in the SLC6A4 gene were used to evaluate the efficiency of the proposed method. All template sequences were retrieved with a 500 bp flanking length (at both sides of SNP) from SNP-Flankplus (http://bio.kuas.edu.tw/snp-flankplus/) [24].

Parameter settings

Four parameters are set for the proposed algorithm, i.e. the number of iterations (generations), the population size, the crossover rate and the mutation rate. The respective values were 1000, 50, 0.6 and 0.001. These values are based on DeJong and Spears' parameter settings [25]. Furthermore, constraints commonly used in natural SNP-RFLP primer design, such as a primer length of between 16~28 nt, a GC% between 20~80%, a primer T_m between 45 and 62°C, a difference of primers T_m of less than 1°C, and a product length larger than 100 bp were used.

Results of the GA-based method

The natural SNP-RFLP primer set results obtained by the GA-based method [14] with the common constraints are shown in Table 1. For the 288 SNPs, all the primer lengths are set between 16 and 28 nt. Table 1 shows that for the parameter settings of DeJong and Spears, 97.6% of the designed primers satisfy the length difference criterion. Most of the primer length differences are between 0 and 5 nt (data not shown). Of the primers, 78.7% primers satisfy GC% the criterion; 86 primers have a GC% of less than 20%, and 22 primers have one higher than 80% (data not shown). The GC clamp criterion is satisfied by 65.9% of the primers. Most of the designed primers also satisfy the T_m and the T_m difference criteria (97.6% and 94.8%). The product length criterion is complied with by 99.9% of the designed primer pairs. Finally, only a few primers (0.2%) do not satisfy the hairpin criterion. Both the dimer and specificity criteria are satisfied.

Results for the proposed method

The natural SNP-RFLP primer set results obtained by the MA-based method with the common constraints are also shown in Table 1. For the 288 SNPs, all the primer lengths are also set between 16 and 28 nt. Table 1 shows that for the parameter settings of DeJong and Spears, 99.2% of the designed primers satisfy the length difference criterion. Most of the primer length differences are between 0 and 5 nt (data not shown). Of the primers, 83.3% primers satisfy GC% the criterion; only 30 primers have a GC% of less than 20%, and 11 primers have one higher than 80% (data not shown). The GC clamp criterion is satisfied by 97.0% of the primers. Most of the designed primers also satisfy the T_m and the T_m difference criteria (97.6% and 97.2%). The product length criterion is complied with by 100% of the designed primer pairs. Finally, only a few primers (0.6%) do not satisfy the hairpin criterion. Both the dimer and specificity criteria are also satisfied like GA-based method.

Comparison of the MA-based and GA-based results

Table 1 shows most compliances of the MA-based method is better than the GA-based method. The compliance with the primer length difference criterion of the MA-based method is 1.6% higher than the GA-based method. The compliance with the GC% criterion using the MA-based method is 4.6% higher than using the GA-based method. The GC clamp compliance of the MA-based method is largely 31.1% higher than the one of the GA-based method. The T_m is not improved when using MA-based method. However, the T_m difference compliances of the MA-based method are 2.4% higher than one of the GA-based method. The product length is lightly 0.1% improved by the MA-based method. Finally, compliances with the hairpin criterion of the MA-based method are 0.4% lower than the GA-based method. The total average fitness is improved by the proposed method.

4 Discussion

In this study, we developed a MA-based natural SNP-RFLP method which has been shown to be a robust search and optimization method than the GA-based method. The *In silico* simulations validated the reliability of the MA-based natural SNP-RFLP primer design method.

Requirements for natural SNP-RFLP primer design

In natural SNP-RFLP primer design, the information about the availability of restriction enzymes is a primary factor. If no any restriction enzymes are available for a given target SNP, the PCR-RFLP experiment is meaningless. In our method, the SNP-RFLPing program [18] is used to mine for restriction enzymes in order to ensure that natural SNP-RFLP primers can be designed. Before designing natural SNP-RFLP primers, a feasible DNA template sequence is required. Hence, the SNP-Flankplus program [24] is introduced to obtain a flanking sequence of 500 bps in length for a given target SNP.

Table 1. Compliance (%) of the GA-based and MA-based methods for SNP-RFLP primer design with the DeJong and Spears parameter settings for SNPs of the SLC6A4 gene. A lower average fitness value is more ideal for designing natural SNP-RFLP primers.

	Constraints									
Method	primer length difference	GC%	GC clamp	T_m	T_m difference	product length	dimer	hairpin	specificity	average fitness
GA	97.6%	78.7%	65.9%	97.6%	94.8%	99.9%	100%	99.8%	100%	4.74
MA	99.2%	83.3%	97.0%	97.6%	97.2%	100%	100%	99.4%	100%	1.17

Primer design constraints concerned

The proposed method investigates the typical primer design constraints, such as primer length, primer length difference, T_m, T_m difference, GC proportion, GC clamp, and PCR product length. Furthermore, secondary structures, such as dimer formation of a primer pair (including cross-dimers and self-dimers), hairpin formation of a primer, and the specificity of primers in a template sequence are also demanded to apply to design natural SNP-RFLP primers.

5 Conclusion

PCR-RFLP is a useful laboratory technique for in small-scale basic research studies of complex genetic diseases that are associated with SNP. This study presents a MA-based method to cooperate with SNP-RFLPing to design the available primers and restriction enzymes effectively. The primer constraints and secondary structures were carefully considered. The SNP-RFLPing is reliable to mine available restriction enzymes for the given target SNP. The flexibility of the proposed method has been demonstrated for the polymorphisms of SLC6A4 gene by *in silico* simulated experiments. The test results shown the proposed method is capable of finding available restriction enzymes and feasible primers for SNP genotyping.

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